

React JS Study Guide

1. What is React?

React is a JavaScript library used for building user interfaces, especially single-page applications.

2. Components

Components are reusable building blocks of a React application. There are mainly two types: Functional Components and Class Components.

3. Props

Props are used to pass data from a parent component to a child component.

4. State

State is used to store data that can change during the lifetime of a component.

5. useState

useState is a React Hook used to create and manage state in functional components.

Example: `const [count, setCount] = useState(0);`

6. useEffect

useEffect is used to perform side effects such as API calls, subscriptions, and updating the document title.

7. useContext

useContext allows components to access shared data without passing props manually through every level.

8. React Router

React Router is used for navigation between different pages in a React application.

9. API Calls

Axios or Fetch API can be used to communicate with a backend server.

10. Virtual DOM

React uses a Virtual DOM to efficiently update the actual DOM when the application state changes.